Higher School of Computer and Digital Sciences and Technologies
file structures and data structures
2 nd year secondary cycle

Tutorial series 2

Exercise 1:

1. Given n observations in p dimensions, what is the training error of the K-Nearest Neighbors (KNN) algorithm (in classification)?
2. Is training a KNN algorithm longer than applying it to label a new observation?
3. Why must the SVM margin be maximized?
4. A SVM with a polynomial kernel was trained on data, and underfitting was observed. Which hyperparameter(s) should be changed, and how?

Exercise 2:

Assume we have a classification problem aiming to determine the class of new instances Xi. The domain of possible class values is {1, 2, 3}.

Based on the following knowledge base, determine the class of instance X6 (3,12,4,7,8) using the K-Nearest Neighbors algorithm with:

- k = 1, then
- k = 3

Instances	A1	A2	A3	A4	A5	Class
X1	3	5	4	6	1	1
X2	4	6	10	3	2	2
X3	8	3	4	2	6	3
X4	2	1	4	3	6	3
X5	2	5	1	4	8	2

Exercise 3:

Suppose you have two classes, A and B, and a new document d to classify. Consider the following training data:

Instances	Class	Cos(v(di), v(d))
d1	A	1
d2	B	0.95
d3	B	0.94
d4	A	0.45
d5	A	0.4
d6	B	0.39

Assuming we use the cosine similarity as the distance metric (i.e., the higher the cosine, the closer the vectors), which class would be assigned to d using a KNN classifier in the following cases:

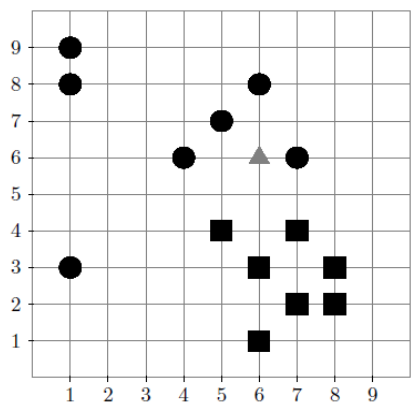
- Majority vote: for k = 3 and k = 5
- Weighted KNN: for k = 3 and k = 5, using the following formula:

Let:

- Sk(d) represent the set of the *k* closest documents to the new document *d*
- v(d) be the vector representation of document *d*
- Ic : Sk(d) → {0,1} be defined for a class *c* and a document *d* as:
    - Ic(dt) = 1 if the class of dt is c
    - Ic(dt) = 0 otherwise

Exercise 4

The 2D feature vectors in the figure below belong to two different classes (circles and rectangles). Classify the object at (6; 6) — in the image represented using a triangle — using k nearest neighbor classification.
Use Manhattan distance (L1 norm) as distance function, and use the non-weighted class counts in the k-nearestneighbor set, i.e. the object is assigned to the majority class within the k nearest neighbors. Perform KNN classification for the following values of k and compare the results with your own "intuitive" result.
K=4, k=7, k=10



Exercise 5 :

Given the following dataset with two points and their respective labels:

Point	x1	x2	Class yi
A	1	2	+1
B	3	4	-1

1. Formulate the dual problem for the Support Vector Machine (SVM) classifier using the given points and labels.
2. Solve for the optimal Lagrange multipliers α1 and α2 using the dual objective function.
3. Compute the weight vector www and the bias term bbb for the optimal hyperplane.
4. Construct the decision function for the classifier.
5. Classify the following new point using the decision function: x=(2,3)